



COUNTER UAS

TECHNOLOGY USA



In conversation with...

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Can you introduce yourself and tell us about your role in advancing counter-UAS capabilities within your organization?

I am a Research Associate with CSIS's Aerospace Security Project. I joined the team to help build and drive our UAS and counter-UAS portfolio. My current work focuses on adversary capabilities and supply chains, domestic counter-drone strategy, and military drone and counter-drone doctrine, with particular emphasis on the Army.

Before CSIS, I served for four years in the Army as a forward observer, including two deployments to Afghanistan. In 2010, while stationed at Camp Casey, South Korea, I participated in a bilateral UAS airborne weapons demonstration. My takeaway then was that the aircraft was impressive but too large and too slow to change how we fought.

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COUNTER UAS

TECHNOLOGY USA

In Afghanistan in 2011 and 2012, a drone, to me, meant an MQ-9 Reaper. I had never seen one in person, much less fought with one overhead. By 2023, when I volunteered as a training advisor in Zaporizhzhia, teaching NATO fire-support procedures to Ukrainian military and territorial defense units, drones ran through nearly everything I taught: how to hide from one, how to call for fire using one, and how to operate with one watching your every move.

Thirteen years separated that demonstration in Korea from my time in Ukraine. I empathize deeply with the horse cavalryman who served long enough to see the tank. That experience informs the questions I bring to CSIS: what changes for the people on the ground, and how quickly can our doctrine, training, and policy catch up?

How do you assess the current drone threat landscape, and what challenges are driving today's counter-UAS priorities?

Much of the excellent work my peers in the UAS field produce focuses on one-way attack drones and swarms. The challenge that most concerns me is persistent battlefield observation: drones have made it much harder to move, resupply, or concentrate forces without being seen.

In Afghanistan, being outside the wire certainly carried its own stresses, but I remember long stretches when I felt reasonably safe despite being deployed to a war zone.

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COUNTER UAS

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In Ukraine, from the moment I arrived in Zaporizhzhia to the moment I returned to Poland, I was never at ease. That feeling came less from the prospect of a weaponized drone striking me than from knowing that being seen by a Russian drone could bring artillery fire.

I am not convinced our counter-drone priorities fully reflect that distinction. An unarmed drone can complete its mission by passing a usable location to a firing unit. Shooting it down afterward may do little for the soldiers it has already exposed.

That makes defeating drone-enabled intelligence, surveillance, and reconnaissance a broader task. It includes concealment, deception, disrupting observation and reporting, and training units to respond when they believe they have been detected. It also requires a shared picture of what is overhead, including friendly drones, and procedures that let people act on that information. I want us to measure success partly by whether a formation can keep operating without being effectively targeted.

What lessons from recent operations and real-world deployments are shaping the future of C-UAS capability development?

I wrote in July about the U.S. counter-drone effort surrounding the World Cup. I was impressed by its scale and

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COUNTER UAS

TECHNOLOGY USA

speed. What concerns me is whether we retain the capability, relationships, and trained personnel assembled for it. A major event ending safely should reinforce the case for preparedness; it can also make that preparedness easier to take for granted.

The alleged plot against the UFC Freedom 250 event at the White House in June illustrates the problem. According to the Justice Department, the conspirators planned to use explosive-armed drones to force an evacuation, then have snipers fire on targets within the fleeing crowd. Their messages discussed raising about \$1,300 for drones and explosive charges. That was their proposed budget, but it shows how little money they believed they needed to add drones to a coordinated attack.

The lesson I take is that drone threats belong in routine security planning for mass gatherings and critical infrastructure, including how an aerial threat changes the response on the ground. That planning cannot depend on an event being prominent enough to warrant a federal task force.

A second lesson is the changing direction of expertise. This spring, Ukraine deployed interceptor drones and specialists to help defend U.S. bases in Jordan following an American request. I trained Ukrainian units on NATO fire-support procedures in 2023; three years later, they are supplying us with operators and techniques. Interoperability now has to account for partners whose combat experience in the drone domain exceeds our own.

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COUNTER UAS

TECHNOLOGY USA

We need a way to absorb what they have learned into our training and procedures while that knowledge is still current.

Which emerging technologies or approaches do you believe will have the greatest impact on detecting, tracking, and defeating UAS threats in the coming years?

Low-cost interceptors are high on my list because affordable rounds let a unit sustain its defense through repeated attacks. Directed energy also deserves serious attention. JIATF-401 has selected five installations for a counter-UAS pilot program incorporating high-energy lasers and high-power microwave systems. That is an opportunity to learn what these technologies can sustain under operational conditions.

My caveat is that cost per engagement tells us only part of the story. The airspace closure at El Paso in February, during a dispute over counter-drone laser use near Fort Bliss, showed how coordination failures can complicate domestic employment. A system's technical performance matters, but so does whether the agencies involved have agreed on how and when it can be used.

The development I would bet on having the greatest impact is whatever reliably shortens the path from detection to an authorized response. Better sensor integration and automated tracking can help, but the unit still needs to know who is responsible, what information is sufficient, and which actions are permitted.

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My reference point is my experience as a forward observer. What makes that role useful is the system around it: training, standardized reporting, communications, and understood responsibilities linking observation to fires. Counter-UAS needs an equivalent connection between detection and action. The observing function may be distributed across people and machines, and the engagement rules will differ by setting. The useful takeaway is that those relationships must be established and rehearsed before a threat appears.

Why is Counter UAS Technology USA such an important forum for the community, and what are you most looking forward to discussing with attendees this year?

Some of the most difficult problems facing the counter-UAS field are organizational, and they do not get solved by people reading each other's white papers. Command architecture is the clearest case: who makes the call, at what echelon, and under what authority? The Pentagon, DHS, the FBI, state and local law enforcement, and allied forces approach those questions under different missions and authorities. Bringing operators, program managers, and policy researchers together helps expose where their assumptions do not line up.

I was pleased to see the Focus Day emphasis on mounted, dismounted, and on-the-move capability, which directly relates to my panel on integrating mobile C-UAS across joint and allied forces. Fixed installation defense has understandably received considerable attention.

Continued...

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Protecting a moving formation introduces additional demands: coverage must move with the unit, communications and power may be constrained, and the counter-UAS equipment has to fit into how that formation fights and sustains itself.

I am most interested in where operational experience meets technology and policy. How are units handling engagement authority when their counter-UAS capability is attached from another organization? Can they maintain a shared air picture while moving? How many trained crews are actually available, as distinct from how many systems have been delivered? And what are allied operators learning from fielding systems such as Merops that have not circulated widely in print?

I expect to leave with a longer list of questions than I arrived with, which is rather the most exciting part of showing up.

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